

Example ① Let $\tau = \min \{t > 0 : W_t = \pm \sqrt{a+bt}\}$ be the first time the BM crosses one of the two parabolic boundaries $\pm \sqrt{a+bt}$, $t \geq 0$, where $a > 0$ and $b \in (0,1)$ are some constants. Compute $E(\tau)$

Sol
 $W_0 = 0, E(W_0^2) = 0 \mid Y_t = W_t^2 - t \mid_{t \geq 0}$ MG (see class notes)

$$0 = E(Y_0) = E(Y_\tau) \quad \text{Using MG stopping time}$$

$$= E(W_\tau^2 - \tau)$$

$$\Rightarrow E(W_\tau^2) - E(\tau) = 0$$

$$\Rightarrow E(a+b\tau) - E(\tau) = 0$$

$$\Rightarrow a + b E(\tau) - E(\tau) = 0$$

$$\Rightarrow E(\tau) = \frac{a}{1-b}$$

② Denote by $\tau = \min \{t > 0 : W_t \leq 2t-4\}$ the first time the BM process crosses the boundary $V_t := 2t-4$, $t \geq 0$.

$$E(\tau), \text{Var}(\tau), E(W_\tau), E(W_\tau^2),$$

$$\text{Laplace transform } l_\tau(s) = E(e^{-s\tau})$$

Sol
 W_t MG, $W_t^2 - t$ MG

$$0 = E(W_0) = E(W_\tau) = E(2\tau-4)$$

$$\Rightarrow 2 E(\tau) - 4 = 0 \Rightarrow E(\tau) = 2$$

1 \setminus \quad 1 \quad 2 \quad \dots

$$|Y_t| := |W_t - t|_{t \geq 0}$$

$$\begin{aligned} 0 &= E(Y_0) = E(Y_\tau) = E(W_\tau^2 - \tau) \\ &= E(W_\tau^2) - E(\tau) \\ &= E((2\tau - 4)^2) - E(\tau) \\ &= E(4\tau^2 + 16 - 16\tau) - E(\tau) \\ &= 4E(\tau^2) + 16 - 32 - 2 \end{aligned}$$

$$W_t \text{ mg} \quad E(\tau^2) = 4.5$$

$$0 = E(W_0) = E(W_\tau)$$

$$W_t^2 - t \text{ mg}$$

$$0 = E(W_0^2 - 0) = E(W_\tau^2 - \tau)$$

$$\Rightarrow E(W_\tau^2) = E(\tau) = 2$$

$$\rightarrow \{Z_t := e^{uW_t - u^2 t/2}\}_{t \geq 0} \text{ mg}$$

$$1 = E(Z_0) = E(Z_\tau) = E(e^{uW_\tau - u^2 \tau/2})$$

$$= E(e^{u(2\tau - 4) - u^2 \tau/2})$$

$$= E(e^{-4u - (\frac{u^2}{2} - 2u)\tau})$$

$$= e^{-4u} E(e^{-s\tau}) \quad , \text{ where } s = \frac{u^2}{2} - 2u$$

$$\Rightarrow E(e^{-s\tau}) = e^{4u}$$

↙ solve

$$u = 2 \pm \sqrt{2(s+2)}$$

$$l_{\tau}(s) = E(e^{-s\tau}) = \exp\{8 - 4\sqrt{2(s+2)}\}$$

Example: Denote by $\tau := \min\{t > 0 : W_t = a$

or $W_t = b\}$ the first time the SBM/Weier process takes one of the values a or b ($a < 0 < b$)

(i) find the dist of W_{τ}

(ii) $E(\tau)$

weel $W_{\tau} \in \{a, b\}$

$$P(W_{\tau} = a) = 1 - P(W_{\tau} = b)$$

$$W_t \text{ m.c.}$$

$$0 = E(W_0) = E(W_{\tau}) = a P(W_{\tau} = a) + b (1 - P(W_{\tau} = a))$$

$$\Rightarrow P(W_{\tau} = a) = \frac{b}{b-a}$$

$$P(W_{\tau} = b) = 1 - \frac{b}{b-a} = \frac{-a}{b-a}$$

$$Y_t := W_t^2 - t \text{ m.c.}$$

$$0 = E(Y_0) = E(Y_{\tau}) = E(W_{\tau}^2 - \tau)$$

$$= E(W_{\tau}^2) - E(\tau)$$

$$E(\tau) = E(W_{\tau}^2)$$

$$= a^2 \times \frac{b}{b-a} - b^2 \times \frac{-a}{b-a}$$

$$= \frac{ab(a-b)}{b-a} = -ab.$$

$$b-a$$